Locating Agents & Rf Values

Exam code: 0620 & 0971

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Help me understand

Locating agents

Extended tier only

- For chromatography to be useful, the chemist needs to be able to see the components move up the paper
  - This is not the case for colourless substances such as amino acids or sugars
- **Locating agents** can be used to see the spots
  - These are substances which react with the sample and produce a visible / coloured spot for the product(s)
- The chromatogram is treated with the agent **after** the chromatography run has been carried out, making the sample runs visible to the naked eye

Retention factor (Rf) values

Extended tier only

- R_f values are used to **identify** the components of mixtures
- The R_f value of a particular compound is always the same
  - However, it does depend on the solvent used
  - If the solvent is changed then the R_f value changes

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

conditions

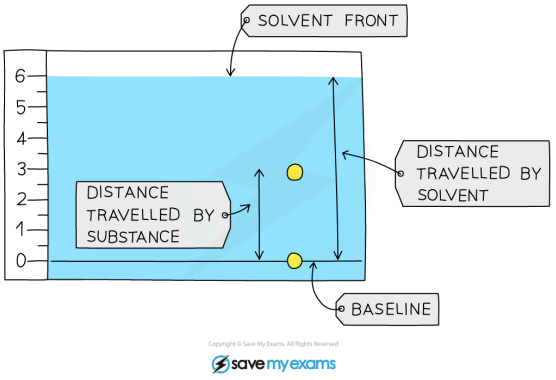
- The retention factor, R_f, is calculated by the equation:

$$R_f = \frac{\text{distance moved by substance}}{\text{distance moved by solvent}}$$

- The R_f value:
  - Is a ratio
  - Has no units
  - Will always be **less** than 1

Worked Example

A student obtained the following chromatogram when carrying out chromatography.



Calculate the R_f value of the substance.

Answer:

- The R_f value of the substances in the chromatogram above can be calculated by:
  - $R_f = \frac{\text{distance moved by substance}}{\text{distance moved by solvent}} = \frac{3}{6} = 0.5$

Examiner Tips and Tricks

When you calculate R_f values in exams, make sure to use your ruler carefully to measure the distance moved by the solvent and the substance, or use any scale provided on the diagram.

Mark schemes can be strict about the values accepted, so it's important to be as accurate as possible when determining these distances.

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